

Currency Converter: Code Walkthrough

A Line-by-Line Technical Deep Dive

ZenCoders

Slide 1: Includes & Struct Definition

```
1 #include <stdio.h>
2 #include <string.h>
3 #include <ctype.h>
4
5 struct Currency {
6     char code[10];
7     char name[50];
8     double rate_to_usd;
9 };
```

Slide 2: Main Function & File Opening

```
1 int main() {  
2  
3     struct Currency currencies[200];  
4     int currency_count = 0;  
5     FILE *file;  
6  
7     file = fopen("rates.txt", "r");
```

Slide 3: File Error Handling (Safety Check)

```
1  if (file == NULL) {  
2      printf("Error: Could not open rates.txt file.\n");  
3      printf("Please make sure the file is in the same folder as the program.\n");  
4      printf("Press Enter to exit.\n");  
5      getchar();  
6      return 1;  
7  }
```

Slide 4: File Parsing Loop (fscanf)

```
1  while(fscanf(file, "%[^,],%[^,],%lf\n",
2      currencies[currency_count].code,
3      currencies[currency_count].name,
4      &currencies[currency_count].rate_to_usd) == 3)
5  {
6      currency_count++;
7      if (currency_count >= 200) {
8          printf("Warning: Hit currency limit. Loaded 200 currencies.\n");
9          break;
10     }
11 }
12
13 fclose(file);
```

Slide 5: Main Menu Loop & Output

```
int choice;

while (1) {
    printf("=====\n");
    printf("        Welcome to the ZenCoders Currency Converter\n");
    printf("        (Loaded %d Currencies from file)\n", currency_count);
    printf("=====\n");
    printf("    1. Convert a Currency\n");
    printf("    2. List all Available Currencies\n");
    printf("    3. Exit Program\n");
    printf("-----\n");
    printf("Enter your choice (1-3): ");
```

Slide 6: Menu Input & 'scanf' Bug Fix

```
1  if (scanf("%d", &choice) != 1) {  
2      while(getchar() != '\n');  
3      printf("\nInvalid input. Please enter a number.\n\n");  
4      continue;  
5  }  
6  printf("\n");  
7  
8  switch (choice) {
```

Slide 7: Case 1 - Setup & Amount Input

```
1      case 1: {
2          char from_code[10], to_code[10];
3          char from_upper[10], to_upper[10];
4          double amount;
5          char again;
6
7          while(1) {
8              printf("Enter amount to convert: ");
9              if (scanf("%lf", &amount) != 1) {
10                 while(getchar() != '\n');
11                 printf("Invalid amount. Please try again.\n");
12                 continue;
13             }
```


Slide 8: Case 1 - String Input & 'toupper'

```
1      printf("Enter 'from' currency code (e.g., PKR): ");
2      scanf("%s", from_code);
3
4      printf("Enter 'to' currency code (e.g., USD): ");
5      scanf("%s", to_code);
6
7      for (int i = 0; i <= strlen(from_code); i++) {
8          from_upper[i] = toupper(from_code[i]);
9      }
10     for (int i = 0; i <= strlen(to_code); i++) {
11         to_upper[i] = toupper(to_code[i]);
12     }
```

Slide 9: Case 1 - Search Algorithm

```
1      double from_rate = 0.0, to_rate = 0.0;
2      int from_found = 0, to_found = 0;
3
4      for(int i = 0; i < currency_count; i++) {
5          if (strcmp(from_upper, currencies[i].code) == 0) {
6              from_rate = currencies[i].rate_to_usd;
7              from_found = 1;
8          }
9          if (strcmp(to_upper, currencies[i].code) == 0) {
10             to_rate = currencies[i].rate_to_usd;
11             to_found = 1;
12         }
13     }
```

Slide 10: Case 1 - Math & Result Output

```
1      printf("\n----- Result ----- \n");
2      if (from_found == 0 || to_found == 0) {
3          printf("Error: Invalid currency code entered.\n");
4      } else {
5          double base_amount = amount / from_rate;
6          double final_amount = base_amount * to_rate;
7
8          printf("%.2f %s = %.2f %s\n", amount, from_upper, final_amount,
9              to_upper);
10     }
11     printf("----- \n\n");
```

Slide 11: Case 1 - "Convert Again?" Loop

```
1      printf("Do another conversion? (y/n): ");
2      scanf(" %c", &again);
3
4      while (again != 'y' && again != 'Y' && again != 'n' && again != 'N') {
5          printf("Invalid option. Please enter 'y' or 'n': ");
6          scanf(" %c", &again);
7      }
8
9      if (again == 'n' || again == 'N') {
10         printf("\n");
11         break;
12     }
13 }
14 break;
15 }
```

Slide 12: Output for 'Convert Currency'

```
Enter amount to convert: 5000
Enter 'from' currency code (e.g., PKR): pkr
Enter 'to' currency code (e.g., USD): eur
```

```
----- Result -----
5000.00 PKR = 15.31 EUR
-----
```

```
Do another conversion? (y/n): y
```

```
Enter amount to convert: 100
Enter 'from' currency code (e.g., PKR): usd
Enter 'to' currency code (e.g., USD): pkr
```

```
----- Result -----
100.00 USD = 28090.00 PKR
-----
```

```
Do another conversion? (y/n): n
```

Slide 13: Case 2 - List Currencies

```
1      case 2: {
2          printf("--- Listing all %d Available Currencies ---\n", currency_count);
3          for (int i = 0; i < currency_count; i++) {
4              printf("    Code: %-10s Name: %s\n", currencies[i].code, currencies[i]
5                  ].name);
6          }
7          printf("-----\n\n");
8          printf("Press Enter to return to the menu...");
9
10         getchar();
11         getchar();
12
13         break;
14     }
```

Slide 14: Output for 'List Currencies'

```
--- Listing all 200 Available Currencies ---
```

```
Code:  USD Name:  US Dollar
```

```
Code:  AED Name:  UAE Dirham
```

```
Code:  AFN Name:  Afghanistan Afghani
```

```
Code:  ALL Name:  Albania Lek
```

```
Code:  AMD Name:  Armenian Dram
```

```
...
```

```
... (and 190+ more)
```

```
...
```

```
Code:  ZAR Name:  South African Rand
```

```
Code:  ZMW Name:  Zambian Kwacha
```

```
-----
```

```
Press Enter to return to the menu...
```

Slide 15: Case 3, Default, & End

```
1         case 3:
2             printf("Thank you for using the ZenCoders Currency converter!\n");
3             return 0;
4
5         default:
6             printf("Invalid choice. Please enter 1, 2, or 3.\n\n");
7             break;
8     }
9 }
10
11 return 0;
12 }
```


Thank You

Questions?